KG COLLEGE OF ARTS AND SCIENCE
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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

PROJECT TITLE

Submitted by

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25BIT107

Under the Guidance of

Trainer Name

Technical Trainer

Technical Department, KG College of Arts and Science

Academic Year: **2025 – 2026**

APRIL 2026

Semester: **I**

DECLARATION

I hereby declare that the project entitled "Scholarship Eligibility Verification and Student Shortlisting System**"** submitted in partial fulfillment of the requirements for the course [B.sc-IT-'B'**]** is a **record of my original work** carried out under the guidance of **[Mugilan]**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

DhikshaR_____

(Your Name)

CERTIFICATE

This is to certify that the project entitled **"Scholarship Eligibility Verification and Student Shortlisting System"** is a **bona fide work** carried out by **[DHIKSHA R] [B.Sc.IT/2025]**, [B.Sc-IT], **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **[Course Name]**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place:

Date:

Submitted for the Viva-Voce Examination held on _____

Guide:
(Technical Trainer, KGCAS)

_____

Head of the Department
(Technical Head, KGCAS)

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ABSTRACT

The Scholarship Eligibility Verification and Student Shortlisting System is a Java-based application designed to automate the process of verifying student eligibility for scholarships and selecting qualified candidates efficiently. In many educational institutions, scholarship selection is done manually, which can lead to errors, delays, and lack of transparency.
This system allows users to input student details such as academic marks, income level, category, and other eligibility criteria. Based on predefined rules, the system automatically verifies eligibility and shortlists deserving candidates.
The application is developed using Core Java concepts such as classes, objects, conditional statements, loops, and exception handling. It can be implemented as a console-based system or a GUI-based application using Java Swing.
The main objective of the system is to reduce manual effort, improve accuracy, and ensure a fair and transparent scholarship selection process.

1. INTRODUCTION

In educational institutions, scholarships play a vital role in supporting students financially. However, the process of verifying eligibility and selecting candidates is often performed manually, which can be time-consuming and error-prone.
Manual systems may lead to:
Incorrect eligibility verification
Bias in selection
Delay in processing applications
Difficulty in managing large student data
To overcome these issues, a computerized system is required.
The Scholarship Eligibility Verification and Student Shortlisting System is designed to automate this process. It evaluates student data based on predefined eligibility criteria and generates a list of qualified candidates.

1.1 Objective

The main objective of this project is to develop a Java-based system that automates scholarship eligibility verification and student shortlisting.
Specific objectives:

* To design a system for managing student data
* To automate eligibility verification based on criteria
* To shortlist students fairly and accurately
* To reduce manual errors
* To provide quick and efficient results
* To demonstrate Core Java concepts

1.2 Overview

The system is a menu-driven application where users can enter student details such as:
Name
Marks
Family income
Category
The system checks whether the student meets eligibility criteria such as:
Minimum marks requirement
Income limit
Reservation category (if applicable)
After verification, the system displays:
Eligible students
Shortlisted candidates

1.3 Features

The **Scholarship Eligibility Verification and Student Shortlisting System** includes the following main features:

Student Data Entry
Eligibility Verification
Automatic Shortlisting
Criteria-Based Filtering
Fast Processing
User-Friendly Interface

2. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. This stage helps in understanding how the system should work and what features are required to solve the problem efficiently.

The analysis phase focuses on identifying the current issues in the existing system and proposing a computerized solution that improves efficiency and accuracy.

2.1 Problem Definition

The current scholarship selection process faces several challenges:
Manual verification is time-consuming
High chances of human error
Lack of transparency
Difficult to handle large number of applications
Hence, an automated system is required to ensure efficient and accurate processing.

2.2 Project Overview

The **Scholarship Eligibility Verification and Student Shortlisting System** is a Java-based application .

This system provides:
Input of student details
Verification using predefined conditions
Automatic shortlisting
Display of eligible candidates
The application simplifies the scholarship process and ensures fairness.

This project is implemented using **Core Java programming concepts** and can be developed as either a **console-based program or a Java Swing GUI application**.

2.3 Scopes and Limitations

Scope

The scope of the **Scholarship Eligibility Verification and Student Shortlisting System** includes the following:

Scope:
* Used in colleges and schools
* Handles student data efficiently
* Automates decision-making

Limitations:
* Works only with predefined criteria
* No database integration (basic version)
* Limited to offline usage

3. SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier.

The **Scholarship Eligibility Verification and Student Shortlisting System** is designed as a menu-driven application where users can select food items, enter quantities, and generate a final bill automatically.

The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function of the system.

3.1 Modules

The main modules of the **Scholarship Eligibility Verification and Student Shortlisting System** are explained below:

1. Student Data Module
Stores student details
Accepts input from user

2. Eligibility Verification Module
Checks conditions like:
Minimum marks (e.g., ≥ 75%)
Income limit (e.g., ≤ ₹2,00,000)
Uses conditional statements

3. Shortlisting Module
Selects eligible students
Prepares final list

4. Display Module
Shows:
Eligible students
Rejected students
Final shortlisted list

5. Exit Module
Terminates the program safely

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4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the **Scholarship Eligibility Verification and Student Shortlisting System**, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

• Problem Analysis
• System Design
• Program Development
• Testing and Validation
• Result and Documentation

Problem Analysis

Identified problems:

Manual processing

Data mismanagement

Delays

Solution:

Develop automated Java system

4.2 System Design

The system is divided into modules:

Input

Processing

Output

4.3 Program Development

Developed using:

Classes and objects

If-else conditions

Loops

Arrays or ArrayLists

4.4 Testing and Validation

Test cases include:

Valid student data → Eligible

Low marks → Not eligible

High income → Rejected

4.5 Result and Documentation

The system successfully:

Verifies eligibility

Shortlists students

Displays results clearly

4.6 Algorithm

Start the program

Enter student details

Check eligibility conditions

If eligible → add to shortlist

Else → reject

Display shortlisted students

End program

4.7. Flowchart

Students should draw a flowchart showing program flow.

Example flow:

Start
↓
Enter Student Data
↓
Check Eligibility

↓
Eligible?
→ Yes → Add to List
→ No → Reject
↓
Display Result
↓
End

Students can draw using:

- draw.io
- Lucidchart
- MS Visio

5. SOURCE CODE

Include important parts of the Java code such as:

- Main class
- Menu logic
- Billing calculation

Example:

```
if(marks >= 75 && income <= 200000){
   System.out.println("Eligible for Scholarship");
} else {
   System.out.println("Not Eligible");
}
```

6. INPUT AND OUTPUT SCREENS

Include screenshots of:
Student data input
Eligibility result
Shortlisted students
DISCUSSION & CONCLUSIO

7. DISCUSSION & CONCLUS

8. REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – *Java: The Complete Reference*

Websites

- https://www.oracle.com/java
- https://www.geeksforgeeks.org
- https://www.w3schools.com
- https://www.javatpoint.com